Curriculum vitae

PERSONAL INFORMATION

MD. FARDIN AL SHAFIK



DAKKHIN KHAILKUR, GACHA, 1704, GAZIPUR, Bangladesh.

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Date of birth 22 September 2003 | Nationality Bangladeshi

EDUCATION

May 2023 – Present

Bachelor of Science in Computer Science & Engineering — EQF 6

IUBAT — International University of Business Agriculture and Technology, Bangladesh

Result: 3.83 / 4.00 | Expected Graduation: May 2027

Principal Subjects: Computer Vision & Image Processing, Linear Algebra, Data Science, Statistics, Numerical Analysis, Applied Calculus, Data Structures & Algorithms, Discrete Mathematics, Software Engineering, Database Management & Systems, Computer Security, Operating System.

2019 – 2021

Higher Secondary Certificate (Science) — EQF 4

Government Science College, Dhaka, Bangladesh

Result: 5.00 / 5.00

EXPERIENCE

May 2025 – Present

Principal Research Team Member

Building Energy & Environmental Management (BEEM) Enhanced by AI, Bangladesh

Interdisciplinary research group developing AI-driven solutions for sustainable energy, environmental monitoring, and occupant comfort optimization in smart cities.

Research Projects

- **University Laboratory Environmental Assessment:** Applied explainable machine learning to assess indoor environmental quality and occupant comfort across 7 university laboratories using 2,000+ environmental samples.
- **Human Behavior & Space-Choice Analytics:** Analyzing occupant comfort preferences and space-selection behavior in educational environments.
- **Computer Vision for Urban Tangled Wire Analysis** Building a multi-label dataset and developing ML models for detection of tangled urban utility wires.

Research Domains Machine Learning • Deep Learning • Data Sciences • Environmental Monitoring

Peer-Reviewed Conference Papers

- **Md. Fardin Al Shafik**, M. A. H. Mukib, M. N. Uddin. *Feature Ranking for Predicting Occupant Thermal Comfort in University Laboratories Using Classification-Based Machine Learning Models for Smart Cities*. AI-Scities 2025, United Kingdom. **DOI:** 10.1051/e3sconf/202673501005
- M. Jakaria, **Md. Fardin Al Shafik**, D. M. Dipta, M. N. Uddin et al. *Predicting Indoor Air Quality in University Laboratories Using Classification-Based Machine Learning Models*. ICCPEAS 2025, Azerbaijan. **DOI:** 10.2991/978-94-6239-668-5_93

Accepted Conference Papers

- M. Mukib, T. U. A. Jengi, M. N. Uddin, **Md. Fardin Al Shafik**, et al. *Visual Comfort Assessment for a Comparative Analysis of Dominant Features Using Machine Learning in Naturally Ventilated Cafeterias*. MedCAVIA'25, Tebessa, Algeria.
- G. M. Rahad, J. Hossain, M. Hasan, M. I. A. Imran, **Md. Fardin Al Shafik**, et al. *Classifying Floor Level Based on Human Comfort and Environmental Parameters Using Machine Learning Approaches for Sustainable Indoor Analytics*. SURE-Built 2026, South Africa.

Undergraduate Thesis

– **ArgMiner: Agent-in-the-Loop or Human-in-the-Loop? A Debate on Post-hoc Feedback for Debates** *Submitted to CHI 2027*
Research focus: Natural Language Processing • Speech Recognition • LLM-based Debate Understanding • Explainable AI • Human–AI Interaction

Conference Abstracts

– M. T. Hasan, T. U. A. Jengi, M. A. H. Mukib, **Md. Fardin Al Shafik**. *Predicting Inhabitant Thermal Comfort in Urban Landscapes Using Machine Learning for Sustainable and Climate-Resilient Design.* MCYR 2025, Czech Republic (Book of Abstracts, p.152).

Conference Poster Presentations

– **Md. Fardin Al Shafik**, M. Jakaria, D. M. Dipta, M. N. Uddin et al. *Predicting Indoor Air Quality in Naturally Ventilated University Cafeterias Using Regression-Based Machine Learning Model.* AGU Annual Meeting 2025, USA. **Poster:** GH43C-VR8762. **Abstract:** ADS

PERSONAL SKILLS

Technical Skills

– **Programming & Data Science:** Python, SQL, NumPy, Pandas, Scikit-learn, Jupyter Notebook, Google Colab
– **Machine Learning:** Supervised Learning, Feature Engineering, Feature Selection, Model Evaluation, SHAP-based Explainability
– **NLP & Speech Processing:** Automatic Speech Recognition (Whisper), Speech/Transcript Processing, LLM-based Text Analysis, Argument Mining
– **Development Tools:** Git/GitHub, Linux, VS Code

EXTRA-CURRICULAR ACTIVITIES

October 2025 - Present

Debate Secretary

Debating Forum of IUBAT (DFI)

– Served as **Event Director** for the *3rd IUBAT Intra Debate Competition 2026*
– Serve as **Safeguarding Committee Member** for *IUBAT-BDF Open 2025*
– Act as **Senior Director** for *DYCMUN Session I (in association with DFI)*

Honours and Awards

– Merit-Based Academic Scholarship (100% Tuition Waiver), IUBAT, Bangladesh
– First Runner-Up, Poster Presentation — IIEC Innovation Competition 2025
– Debater of the Tournament — Freshman Debate Competition (DFI)
– Second Runner-Up, Freshman Mathematics Olympiad — IUBAT Mathmatics Society (IMS)

REFERENCES

Research Supervisor (BEEM)

Mohammad Nyme Uddin, PhD

Assistant Professor
Department of Architectural Engineering, College of Engineering,
Imam Mohammad Ibn Saud Islamic University (IMSIU), Saudi Arabia
Email: mnuddin@imamu.edu.sa Phone: +966 567 397 295

Thesis Supervisor

Saha Kuljit Shantanu

Lecturer
Department of Computer Science and Engineering,
IUBAT - International University of Business Agriculture and Technology, Bangladesh
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DECLARATION

I declare that the information provided in this CV is true and accurate.



Md. Fardin Al Shafik
October 5, 2026